Problems marked with a (*) are a bit more complex.

8.1. Jordan-null set. Let $X \subset \mathbb{R}$ be a Jordan-null set (as in Definition 13.8).

(a) Show rigorously that $X \times X \subset \mathbb{R}^2$ is also Jordan-null.

(b) Show rigorously that $X \times [0,1] \subset \mathbb{R}^2$ is also Jordan-null.

8.2. True or False.

1. A bounded countable set is always Jordan-null.
2. A countable set is always Lebesgue-null.
3. Let $D \subset [0,1]$ be a dense set (i.e., $\overline{D} = [0,1]$). Then $\mu_{out}(D) = 1$. (μ_{out} was defined in Definition 13.7).
4. Let $X, Y \subset [0,1]$ Jordan measurable sets such that $\mu(X) > 1/2$ and $\mu(Y) > 1/2$. Then $X \cap Y \neq \emptyset$.
5. Let $X, Y \subset [0,1]$ such that $\mu_{out}(X) > 1/2$ and $\mu_{out}(Y) > 1/2$. Then $X \cap Y \neq \emptyset$.

8.3. Fat Boundary. Construct an open subset $U \subset \mathbb{R}$ for which the boundary ∂U is not a null set.

8.4. Multiple Choice. Let $U \subset \mathbb{R}^n$ be a nonempty, open subset, $f\colon U \to \mathbb{R}^m$ a function, and $N \subset U$ a Jordan null set. In which of the following cases is the image $f(N) \subset \mathbb{R}^m$ necessarily a null set? Attention: Only one answer is correct!

1. If f is uniformly continuous.
2. If f is uniformly continuous and $m \geq n$.
3. If f is locally Lipschitz continuous.
4. If f is locally Lipschitz continuous and $m \geq n$.

8.5. Change of variables and Jacobians. For each of the following domains and change of variables find the jacobian and the appropriate transformed domain. There is no need to actually compute the integrals!

1. $A := \{(x_1, x_2) \in \mathbb{R}^2 : x_1 > 0, x_2 < x_1, 1 < x_1^2 + x_2^2 < 4\}$ and $x_1 = r\cos\theta, x_2 = r\sin\theta$. Using the change of variables formula, complete the dots in the following formula

$$\int_A x_1^2 \sin(x_2)\, dx_1 dx_2 = \int_{\dots} \dots\, dr d\theta$$

2. $B := \{(x,y) \in \mathbb{R}^2 \,|\, 1 < xy < 2, x^2 < y < 2x^2\}$ and $u := xy, v := x^2$. Using the change of variables formula, complete the dots in the following formula

$$\int_B y^2 e^{-xy} dx dy = \int_{\dots} \dots\, du dv$$

3. $C := \{(x, y, z) \in \mathbb{R}^3 \,|\, 1 < z - 2y < 2, 0 < z < 1, -2 < 3x + y + z < 0\}$ and $u := z, v := z - 2y, w := 3x + y + z$. Using the change of variables formula, complete the dots in the following formula

$$\int_C xyz \; dxdy = \int_{\dots} \dots \, dudvdw.$$

8.6. The Cantor set. (*) Let $X \subset [0, 1]$ be the set of all real numbers whose decimal expansion does not contain the digit 8.[1] Show that:

1. X is a Lebesgue null set,
2. X is uncountable,
3. $X \times X \subset [0, 1]^2$ is a Lebesgue null set.
4. Show that X is compact (it is important the choice made in the footnote!).

[1]The decimal expansion is not always unique. For example, $0.8 = 0.79999\ldots$. Whenever x has at least one decimal expansion not containing 8, we rule that x **belongs to** X, so for example $0.3257\overline{9} \in X$, $0.3258\overline{9} \in X$

Hints:

8.2.5 μ_{out} can be positive and large on very sparse sets...

8.3 Any open subset of $\mathbb{R}$ is an union of disjoint open intervals. Try to achieve that U has a very small "total volume", but still contains all rational numbers in $[0, 1]$.